RS* DESCENT FOR THE COMPLEXES STACK ALONG PUSHOUTS ALONG THICKENINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-complexes (canonical tag 0DLE). cited at statement line 5666. Standing situation.
- Lemma more-morphisms-lemma-pushout-along-thickening (canonical tag 07RT). cited at statement lines 5674-5675. The pushout square exists.
- Lemma spaces-pushouts-lemma-pushout-along-thickening-schemes (canonical tag 07SY). cited at proof lines 5692-5693 and 5723-5724. Base bijection; alternative route for bijectivity.
- Remark remark-complexes-base-change (canonical tag 0DLI). cited at proof lines 5705-5706. Base change reduction to B = Y'.
- Lemma artin-lemma-pushout (canonical tag 07WN). cited at proof lines 5721-5722. Bijectivity of Isom over the pushout for algebraic spaces.
- Lemma spaces-pushouts-lemma-equivalence-categories-spaces-pushout-flat (canonical tag 07W3). cited at proof lines 5747-5748. X_Z, X_Y, X_{Z'}, X_{Y'} form a pushout square.
- Lemma spaces-pushouts-lemma-pushout-along-thickening-derived (canonical tag 0DL7). cited at proof lines 5749-5750. Descent of the derived object E_{Y'}.
- Lemma lemma-complexes (canonical tag 0DLD). cited at proof lines 5756-5757. Condition (2) is a property on points.
- Lemma spaces-more-morphisms-lemma-thickening-relatively-perfect (canonical tag 0DL4). cited at proof lines 5759-5760. E_{Y'} is Y'-perfect.

Source reference. Original definition 0DLE: sources/stacks-chow/tags.txt, source lines 17044–17044. Cited at statement line 5666: the lemma is phrased in this situation; load-bearing.

Source reference. Original lemma 07RT: sources/stacks-chow/tags.txt, source lines 9878–9878. Cited at statement lines 5674-5675: existence of the pushout square; load-bearing.

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Source reference. Original lemma 07SY: sources/stacks-chow/tags.txt, source lines 9918–9918. Cited at proof lines 5692-5693 (base bijection) and 5723-5724 (alternative for bijectivity); load-bearing twice.

Source reference. Original remark 0DLI: sources/stacks-chow/tags.txt, source lines 17048–17048. Cited at proof lines 5705-5706: replace B by Y' and X by X x_B Y'; load-bearing for the reduction.

Source reference. Original lemma 07WN: sources/stacks-chow/tags.txt, source lines 10048–10048. Cited at proof lines 5721-5722: bijectivity of the Isom map for algebraic spaces; load-bearing for full faithfulness.

Source reference. Original lemma 07W3: sources/stacks-chow/tags.txt, source lines 10028–10028. Cited at proof lines 5747-5748: X_Z -> X_Y affine and X_Z -> X_{Z'} thickening pushout; load-bearing.

Source reference. Original lemma 0DL7: sources/stacks-chow/tags.txt, source lines 17037–17037. Cited at proof lines 5749-5750: existence of E_{Y'} with compatible isomorphisms; load-bearing.

Source reference. Original lemma 0DLD: sources/stacks-chow/tags.txt, source lines 17043–17043. Cited at proof lines 5756-5757: condition (2) is a property on points; load-bearing.

Source reference. Original lemma 0DL4: sources/stacks-chow/tags.txt, source lines 17034–17034. Cited at proof lines 5759-5760: E_{Y'} is Y'-perfect; load-bearing.

Lemma 0.2. *In Situation 0DLE. Let*

$$\begin{array}{ccc} Z & \longrightarrow & Z' \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y' \end{array}$$

be a pushout in the category of schemes over S where $Z \to Z'$ is a finite order thickening and $Z \to Y$ is affine, see More on Morphisms, Lemma 07RT. Then the functor on fibre categories

$$\mathcal{C}omplexes_{X/B,Y'} \longrightarrow \mathcal{C}omplexes_{X/B,Y} \times_{\mathcal{C}omplexes_{X/B,Z}} \mathcal{C}omplexes_{X/B,Z'}$$

is an equivalence.

Proof. Observe that the corresponding map

$$B(Y') \longrightarrow B(Y) \times_{B(Z)} B(Z')$$

is a bijection, see Pushouts of Spaces, Lemma 07SY. Thus using the commutative diagram

$$\begin{array}{ccc} \mathcal{C}omplexes_{X/B,Y'} & \longrightarrow & \mathcal{C}omplexes_{X/B,Y} \times_{\mathcal{C}omplexes_{X/B,Z}} \mathcal{C}omplexes_{X/B,Z'} \\ \downarrow & & \downarrow \\ B(Y') & \longrightarrow & B(Y) \times_{B(Z)} B(Z') \end{array}$$

we see that we may assume that Y' is a scheme over B'. By Remark 0DLI we may replace B by Y' and X by $X \times_B Y'$. Thus we may assume $B = Y'$.

Assume $B = Y'$. We first prove fully faithfulness of our functor. To do this, let ξ_1, ξ_2 be two objects of $\mathit{Complexes}_{X/B}$ over Y'. Then we have to show that

$$\mathit{Isom}(\xi_1, \xi_2)(Y') \longrightarrow \mathit{Isom}(\xi_1, \xi_2)(Y) \times_{\mathit{Isom}(\xi_1, \xi_2)(Z)} \mathit{Isom}(\xi_1, \xi_2)(Z')$$

is bijective. However, we already know that $\mathit{Isom}(\xi_1, \xi_2)$ is an algebraic space over $B = Y'$. Thus this bijectivity follows from Artin's Axioms, Lemma 07WN (or the aforementioned Pushouts of Spaces, Lemma 07SY).

Essential surjectivity. Let $(E_Y, E_{Z'}, \alpha)$ be a triple, where $E_Y \in D(\mathcal{O}_Y)$ and $E_{Z'} \in D(\mathcal{O}_{X_{Z'}})$ are objects such that $(Y, Y \to B, E_Y)$ is an object of $\mathcal{C}\mathit{omplexes}_{X/B}$ over Y, such that $(Z', Z' \to B, E_{Z'})$ is an object of $\mathcal{C}\mathit{omplexes}_{X/B}$ over Z', and $\alpha : L(X_Z \to X_Y)^* E_Y \to L(X_Z \to X_{Z'})^* E_{Z'}$ is an isomorphism in $D(\mathcal{O}_{Z'})$. That is to say

$$((Y, Y \to B, E_Y), (Z', Z' \to B, E_{Z'}), \alpha)$$

is an object of the target of the arrow of our lemma. Observe that the diagram

$$\begin{array}{ccc} X_Z & \longrightarrow & X_{Z'} \\ \downarrow & & \downarrow \\ X_Y & \longrightarrow & X_{Y'} \end{array}$$

is a pushout with $X_Z \to X_Y$ affine and $X_Z \to X_{Z'}$ a thickening (see Pushouts of Spaces, Lemma 07W3). Hence by Pushouts of Spaces, Lemma 0DL7 we find an object $E_{Y'} \in D(\mathcal{O}_{X_{Y'}})$ together with isomorphisms $L(X_Y \to X_{Y'})^* E_{Y'} \to E_Y$ and $L(X_{Z'} \to X_{Y'})^* E_{Y'} \to E_Z$ compatible with α. Clearly, if we show that $E_{Y'}$ is Y'-perfect, then we are done, because property (2) of Lemma 0DLD is a property on points (and Y and Y' have the same points). This follows from More on Morphisms of Spaces, Lemma 0DL4. $\square$